Introduction to Affine Grassmannian

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1 Definition of Affine Grassmannian

We will always work over $k = \overline{k}$ and G/k a connected reductive group.

Let's start by defining $\mathrm{Gr}_{\mathrm{GL}_n}$. Let V be the standard representation of $G = \mathrm{GL}_n$, and let $F \coloneqq k((t))$, and $\mathcal{O} \coloneqq k[[t]]$. A *lattice* in $V \otimes_k F$ is a (automatically free) $\mathcal{O}$-submodule of rank n. Thus any lattice is the $\mathcal{O}$-span of some linearly independent $v_1, \cdots, v_n \in V \otimes_k F$. The k-points of the affine Grassmannian for GL_n are the lattices in $V \otimes_k F$:

$$\mathrm{Gr}_{\mathrm{GL}_n}(k) = \{\text{lattices in } V \otimes_k F\}$$

Notice that this set has a transitive $\mathrm{GL}_n(F)$-action. Let's fix the standard lattice $\Lambda_0 \coloneqq V \otimes \mathcal{O}$, which has stablizer $\mathrm{GL}_n(\mathcal{O})$. It follows that:

$$\mathrm{Gr}_{\mathrm{GL}_n}(k) \simeq \mathrm{GL}_n(F)/\mathrm{GL}_n(\mathcal{O})$$

*based on lectures by Zhiwei Yun

We will eventually endow $\mathrm{Gr}_{\mathrm{GL}_n}$ with the structure of an ind-scheme. First, we would like to write down its functor of points; for a k-algebra R, we expect that an R-point of $\mathrm{Gr}_{\mathrm{GL}_n}$ to be a "family of lattices in $V \otimes F$ over Spec R." To motivate the right such notion, we rephrase our current definition of a lattice. Fix a basis $e_1, \cdots, e_n$ of V. Then the completion of the span of

$$\{t^i e_j \mid i \in \mathbb{Z}, 1 \leq j \leq n\}$$

is $V \otimes F((t))$. It is easy to see that if $\Lambda \subset V \otimes_k F$ is a lattice, then there exists some N such that $t^N \Lambda \subset \Lambda_0$. Conversely, any $\mathcal{O}$-submodule between $t^N \Lambda_0 \subset t^{-N} \Lambda_0$ is automatically a rank n lattice.

Therefore, we should define a lattice in $V \otimes_k R((t))$ to be an $R[[t]]$-module $\Lambda \subset V \otimes_k R((t))$ such that:

1. There exists $N \in \mathbb{Z}_{>0}$ such that $t^N \Lambda_{0,R} \subset \Lambda \subset t^{-N} \Lambda_{0,R}$;
2. $t^{-N} \Lambda_{0,R} / \Lambda$ is locally free as an R-module.

Here $\Lambda_{0,R} \coloneqq V \otimes R[[t]]$. We now define a functor:

$$\mathrm{Gr}_{\mathrm{GL}_n}(R) \coloneqq \{\text{lattices in } V \otimes_k R((t))\}$$

Since we want to do geometry, so we wish that $\mathrm{Gr}_{\mathrm{GL}_n}$ is not merely a stack. While $\mathrm{Gr}_{\mathrm{GL}_n}$ is not strictly a scheme, it has truncations:

$${}^N\mathrm{Gr}_{\mathrm{GL}_n} : R \mapsto \{\text{lattices } \Lambda : t^N \Lambda_{0,R} \subset \Lambda \subset t^{-N} \Lambda_{0,R}\}$$

This is in fact a projective scheme. To see this, notice that such Λ are in bijection with projective R-submodules of $t^{-N} \Lambda_{0,R} / t^N \Lambda_{0,R}$ stable under multiplication-by-t. Taking the union over its rank i, we obtain a closed embedding

$${}^N\mathrm{Gr}_{\mathrm{GL}_n} \hookrightarrow \coprod_i \mathrm{Gr}\binom{t^{-N}\Lambda_0 / t^N \Lambda_0}{i}$$

Thus $\mathrm{Gr}_{\mathrm{GL}_n} = \bigcup_N {}^N\mathrm{Gr}_{\mathrm{GL}_n}$ is an *ind-scheme*.

Remark. *For general R, the action of $\mathrm{GL}(R((t)))$ on $\mathrm{Gr}_{\mathrm{GL}_n}(R)$ is not transitive. However, it is transitive in the flat (fpqc) topology. Moreover, the stablizer of $\Lambda_{0,R}$ is still $\mathrm{GL}_n(R[[t]])$. Thus $\mathrm{Gr}_{\mathrm{GL}_n}$ is the sheafification of*

$$R \mapsto \mathrm{GL}_n(R((t)))/\mathrm{GL}_n(R[[t]])$$

in the flat definition.

This motivates:

Definition 1.1. Let G be a connected reductive group. The affine Grassmanian Gr_G is the flat sheafification of

$$R \mapsto G(R((t)))/G(R[[t]])$$

Concretely, given a faithful representation $G \hookrightarrow \mathrm{GL}_n$, we get a map $\mathrm{Gr}_G \hookrightarrow \mathrm{Gr}_{\mathrm{GL}_n}$. So we should view Gr_G is lattices with extra conditions.

Example. *Suppose $G = \mathrm{SL}_n \subset \mathrm{GL}_n$. We will show that $\mathrm{Gr}_{\mathrm{SL}_n} = (\mathrm{Gr}_{\mathrm{GL}_n})^0$ is the* neutral component *of $\mathrm{Gr}_{\mathrm{GL}_n}$. We write*

$$\mathrm{Gr}_{\mathrm{GL}_n} = \coprod_{i \in \mathbb{Z}} \mathrm{Gr}^{(i)}_{\mathrm{GL}_n}$$

where $\mathrm{Gr}^{(i)}_{\mathrm{GL}_n}$ parametrizes lattices Λ whose image in $t^{-N}\Lambda_{0,R}/t^N\Lambda_{0,R}$ has R-rank $Nn + i$ (this condition does not depend on N). For example $\Lambda_0 \in \mathrm{Gr}^{(0)}_{\mathrm{GL}_n}$; if we take $\Lambda = \mathrm{Span}_{\mathcal{O}}(te_1, t^{-1}e_2, e_3, te_4)$, then $t\Lambda_0 \subset \Lambda \subset t^{-1}\Lambda$ and $\Lambda/t\Lambda_0$ has rank $3 = 1 \cdot 4 - 1 \implies \Lambda \in \mathrm{Gr}^{(-1)}_{\mathrm{GL}_n}$. The result follows from the fact that $\mathrm{SL}_n(F)$ acts transitively on each $\mathrm{Gr}^{(i)}_{\mathrm{GL}_n}(k)$.

Example. *Suppose $G = \mathrm{Sp}_{2n} = \mathrm{Sp}(V, \langle -, - \rangle)$. Then*

$$\mathrm{Gr}_G(R) = \{\Lambda \text{ lattices in } V \otimes_k R((t)) : \langle -, - \rangle|_\Lambda \text{ is a perfect } R[[t]]\text{-valued pairing}\}$$

Let us elaborate on the perfect pairing condition. First extend $\langle -, - \rangle$ to a symplectic form (over $R((t))$) on $V \otimes_k R((t))$ by linearity. Set $M_N \coloneqq t^{-N}\Lambda_{0,R}/t^N\Lambda_{0,R}$ and consider the R-bilinear symplectic pairing:

$$(-, -) : M_N \times M_N \to R$$

$$(\overline{v_1}, \overline{v_2}) \mapsto \mathrm{Res}_{t=0}(\langle v_1, v_2 \rangle dt) = \text{coefficient of } t^{-1} \text{ in } \langle v_1, v_2 \rangle$$

It is easy to check this is well-defined and perfect. In fact $\langle -, - \rangle|_\Lambda$ is perfect iff $\Lambda/t^N\Lambda_{0,R} \subset M_N$ is a Lagrangian subspace for appropriately chosen N. So we can write:

$${}^N\mathrm{Gr}_{\mathrm{Sp}_{2n}}(R) = \{\text{lattices } t^N\Lambda_{0,R} \subset \Lambda \subset t^{-N}\Lambda_{0,R} : \Lambda/t^N\Lambda_{0,R} \subset M_N \text{ is Lagrangian for } (-, -)\}$$

2 Loop Space

Let $G/\mathbb{C}$ be a complex reductive group with maximal compact K. We will describe Gr_G in terms of K.

Definition 2.1. The smooth based loop space is defined to be:

$$\Omega_{\mathrm{sm}}(K) \coloneqq \{\gamma : S^1 \to K \text{ smooth} : \gamma(1) = 1\}$$

After fixing an embedding $K \subset U(n)$, then polynomial based loops space is defined to be:

$$\Omega_{\mathrm{pol}}K \coloneqq \{\gamma \in \Omega_{\mathrm{sm}}K : \text{each } \gamma(z)_{ij} \text{ is a finite Fourier series in } z\}$$

Note this deos not depend on the choice of embedding $K \subset U(n)$.

We will construct a map $\Omega_{\mathrm{pol}}K \to \mathrm{Gr}_G(\mathbb{C})$ that will turn out to be a homoemorphism (for the topologies induced by polynomial degree and by the inductive limit respectively). For simplicity, we first consider the case of GL_n. Then any polynomial loop

$$\gamma : S^1 \to U(n)$$

has form $\gamma(z) = \sum_i A_i z^i$ a finite sum with coefficients $A_i \in \mathfrak{gl}_n(\mathbb{C})$. In particular, we can evaluate γ on any $z \in \mathbb{C}^\times$ to get a matrix.

Lemma 2.2. *$\gamma(\overline{z}^{-1}) = (\gamma(z)^*)^{-1}$, where $(-)^*$ is the Hermitian adjoint of a matrix.*

Proof. Clearly, the claim holds on S^1. Then we deduce it's true everywhere since both sides are antiholomorphic. □

A consequence is that $\gamma(z) \in \mathrm{GL}_n(\mathbb{C})$ (in priori we don't know this), so from γ we constructed a regular(algebraic) map:

$$g_\gamma : \mathbb{C}^\times \to \mathrm{GL}_n(\mathbb{C}) \iff g_\gamma \in \mathrm{GL}_n(\mathbb{C}[t, t^{-1}])$$

Composing with $\mathrm{GL}_n(\mathbb{C}[t, t^{-1}]) \subset \mathrm{GL}_n(\mathbb{C}((t))) \to \mathrm{Gr}_{\mathrm{GL}_n}(\mathbb{C})$ gives the desired map:

$$\Omega_{\mathrm{pol}} U(n) \to \mathrm{Gr}_{\mathrm{GL}_n}(\mathbb{C})$$

$$\gamma \mapsto g_\gamma \cdot \Lambda_0$$

In general, we may assume that we have an emebding $G \hookrightarrow \mathrm{GL}_n$ sending $K \hookrightarrow U(n)$. Then the same argument as above works, except that our lemma should instead interwine the Cartan involution $\theta : z \mapsto \overline{z}^{-1}$ for S^1 with the Cartan involution $\sigma : G \to G$ for G. Now these two involutions induce a new involution on $G(\mathbb{C}[t, t^{-1}])$ given by:

$$\Theta : g(t) \mapsto \sigma(g(\theta(t)))$$

and we get a sequence of maps:

$$\Omega_{\mathrm{pol}} K \to G(\mathbb{C}[t, t^{-1}])^\Theta / K \to G(\mathbb{C}[t, t^{-1}])/G(\mathbb{C}[t]) \simeq \mathrm{Gr}_G$$

It terms out that these are all homoemorphisms.

Since we can approximate all continuous functions by polynomials, we can show that $\Omega_{\mathrm{lop}} K$ is homotopy equivalent to the based loop sapce ΩK. This allos us to compute the (co)homology and homotopy groups of $\mathrm{Gr}_G(\mathbb{C})$. For example, we know:

$$\pi_0 \mathrm{Gr}_G(\mathbb{C}) \simeq \pi_0 \Omega_K \simeq \pi_1 K \simeq \pi_1^{\mathrm{alg}} G$$

(The first isomorphism involves some Tate twist). There is also a canonical isomorphism:

$$\pi_0 \mathrm{Gr}_G \simeq X_*(T)/(\text{coroot lattice}) = X_*(T)/Q^\vee$$

Example. *The cocharacter lattice of PGL_n is $\mathbb{Z}^n/\Delta\mathbb{Z}$, and its coroot lattice is spanned by $(e_i - e_j)$. Thus we have an isomorphism:*

$$X_*/Q^\vee \simeq \mathbb{Z}^n/(\Delta\mathbb{Z}, e_i - e_j) \xrightarrow{\sim} \mathbb{Z}/n\mathbb{Z}$$

where the second isomorphism is given by summation. On ther other hand, we have:

$$\mathrm{Gr}_{\mathrm{PGL}_n} \simeq \mathrm{Gr}_{\mathrm{GL}_n}/t^{\mathbb{Z}}$$

namely we identify the lattices Λ and $t\Lambda$. Notice that this identifies the components $\mathrm{Gr}_{\mathrm{GL}_n}^{(i)}$ and $\mathrm{Gr}_{\mathrm{GL}_n}^{(i+n)}$. So the group of connected components is indeed $\mathbb{Z}/n\mathbb{Z}$.

Remark. *Canonically, we have:*

$$\pi_1^{\mathrm{alg}}(\mathrm{PGL}_n) \simeq \mu_n$$

The twist occurs because our map $\Omega_{\mathrm{pol}}K \to \mathrm{Gr}_G$ *exploits the structure of* $S^1 \subset \mathbb{C}^\times$ *in a way relying on the analytic topology of* $\mathbb{C}$*.*

The next goal is to understand (co)homology. Still as topological spaces:

$$\Omega_{\mathrm{pol}}K \simeq \mathrm{Gr}_G(\mathbb{C})$$

This endows $\mathrm{Gr}_G(\mathbb{C})$ with a group structure coming from pointwise multiplication of loops in $\Omega_{\mathrm{pol}}K$, although this group structure is real algebraic (not complex algebraic). This means that the (co)homology of $\mathrm{Gr}_G(\mathbb{C})$ has the structure of a Hopf algebra.

To understand the rational cohomology of ΩK and its neutral component $(\Omega K)^\circ$, we consider the fibre sequence:

$$\Omega K \to \mathrm{pt} \to K$$

$$K \to \mathrm{pt} \to BK$$

We have

$$H^*(BK, \mathbb{Q}) \simeq \mathbb{Q}[f_1, \cdots, f_r]$$

where f_i are homogeneous generators of degree $2d_i$ for d_i the fundamental degrees of K. This comes from the isomorphism:

$$H^*(BK, \mathbb{Q}) \simeq H^*(BT_K, \mathbb{Q})^W$$

Now let's assume G is semisimple, so that $d_i \geq 2$. In this case, we have:

$$H^*((\Omega K)^\circ, \mathbb{Q}) \simeq \mathbb{Q}[e_1, \cdots, e_r]$$

for e_i are homogeneous of degree $2d_i - 2$. The e_i are obtained from f_i via transgression. Namely, we pull back f_i to a class in $\Sigma^2\Omega^2 BK$ via adjunction then integrate in the S^2-direction to obtain e_i. It turns out that we may choose e_i to be primitive with respect to the Hopf algebra structure on $H^*((\Omega K)^\circ)$.

3 Stratification of Affine Grassmannian

In this section, we let $F = k((t))$ and $\mathcal{O} = k[[t]]$. We will discuss various stratifciations of Gr_G.

Let's start from **Cartan decomposition**. At the level of k-points, consider the left $G(\mathcal{O})$-action on $\mathrm{Gr}_G(k) = G(F)/G(\mathcal{O})$. Choose a maximal torus $T \subset G$ and let W be the Weyl group. Given a cocharacter $\lambda : \mathbb{G}_m \to T$, denote the image of $t \in F^\times$ in $T(F)$ by t^λ.

Theorem 3.1 (Cartan decomposition)**.** *We have decomposition:*

$$G(F) = \coprod_{\lambda \in X_*(T)/W} G(\mathcal{O}) t^\lambda G(\mathcal{O})$$

Proof. (sketched) We will prove:

$$\mathrm{Gr}_G(k) = \coprod_{\lambda \in X_*(T)/W} G(\mathcal{O}) \cdot t^\lambda$$

Let us just prove that each $G(\mathcal{O})$-orbit contains some t^λ. The key is to consider the 1-dimensional torus $\mathbb{G}_m^{\mathrm{rot}}(k)$ acting on $\mathrm{Gr}_G(k)$ by *loop rotation*: for $s \in \mathbb{G}_m^{\mathrm{rot}}$ and $g = g(t) \in \mathrm{Gr}_G$, the action is $s \cdot g(t) = g(st)$.

The first step is to compute the fixed points of this action. These turn out to be exactly:

$$\coprod_{\lambda \in X_*(T)/W} G(k) \cdot t^\lambda$$

To see this, embed $G \hookrightarrow \mathrm{GL}_n$ to obtain a $\mathbb{G}_m^{\mathrm{rot}}$-equivariant embedding $\mathrm{Gr}_G \hookrightarrow \mathrm{Gr}_{\mathrm{GL}_n}$. Then it suffices to compute the fixed points in $\mathrm{Gr}_{\mathrm{GL}_n}$.

The next step is to show that every $g(t)$ has a limit point under the $\mathbb{G}_m^{\mathrm{rot}}$-action, namely:

$$\lim_{s \to 0} s \cdot g(t) \in \mathrm{Gr}_G$$

exists. The reason for the existence is that Gr_G is the union of projective varieties with a $\mathbb{G}_m^{\mathrm{rot}}$-action, and a torus action on projective space always has limit points.

The last step is to show that if $\lim_{s \to 0} s \cdot g(t) = t^\lambda$ up to $G(k)$-translation, then $g(t) \in G(\mathcal{O})t^\lambda$. To do this, we work in a neighborhood around t^λ and study the infinitesimal action using local coordinates given by the affine root spaces in $\mathfrak{g}(F)$. □

The Cartan decomposition allows us to talk about relative position in Gr_G given by dominant coweight. For GL_n, two lattices (Λ, Λ') are in relative position: $\lambda = (\lambda_1, \cdots, \lambda_n)$ with $\lambda_1 \geq \cdots \geq \lambda_n$ iff there exists a basis $v_1, \cdots, v_n$ of F^n such that

$$\Lambda = \mathrm{Span}_{\mathcal{O}}(v_1, \cdots, v_n), \quad \Lambda' = \mathrm{Span}_{\mathcal{O}}(t^{\lambda_1} v_1, \cdots t^{\lambda_n} v_n)$$

In particular, for GL_n we can consider:

$$\mathrm{Gr}^+_{\mathrm{GL}_n} = \coprod_{\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq \mathbf{0}} \mathrm{GL}_n(\mathcal{O}) \begin{pmatrix} t^{\lambda_1} & & \\ & \cdots & \\ & & t^{\lambda_n} \end{pmatrix} \mathrm{GL}_n(\mathcal{O})/\mathrm{GL}_n(\mathcal{O}) = \{\Lambda \subset \mathcal{O}^n\}$$

Recall that the connected components of $\mathrm{Gr}_{\mathrm{GL}_n}$ parametrize lattices Λ whose relative dimension with the standard lattice $\mathcal{O}^n$ is fixed. The intersection of $\mathrm{Gr}^+_{\mathrm{GL}_n}$ with each connected component is finite-dimenisonal and increasing in dimension. For example:

$$\mathrm{Gr}^+_{\mathrm{GL}_n} \cap \mathrm{Gr}^{(0)}_{\mathrm{GL}_n} = \{\mathcal{O}^n\}$$

$$\mathrm{Gr}^+_{\mathrm{GL}_n} \cap \mathrm{Gr}^{(1)}_{\mathrm{GL}_n} \simeq \mathbb{P}^{n-1}$$

$$\dim \mathrm{Gr}^+_{\mathrm{GL}_n} \cap \mathrm{Gr}^{(2)}_{\mathrm{GL}_n} = 2(n-1)$$

Which $\mathrm{GL}_n(\mathcal{O})$-orbit does a given $\Lambda \subset \mathcal{O}^n$ lie in? By the structure theorem for finitely generated modules over a PID, the torsion $\mathcal{O}$-module $\mathcal{O}^n/\Lambda$ is a sum of cyclic modules:

$$\mathcal{O}^n/\Lambda \simeq \mathcal{O}/t^{\mu_1} \oplus \cdots \oplus \mathcal{O}/t^{\mu^r}$$

for μ^i increasing. Then the dominant coweight indexing the orbit of Λ is

$$\lambda = (\mu_r, \cdots, \mu_1, 0, \cdots, 0)$$

We choose a Borel $B \supset T$, so we can talk about dominant coweights $X_*(T)^+$.

Definition 3.2. For a dominant coweight λ, let $\mathrm{Gr}_\lambda = G(\mathcal{O})t^\lambda$ and $\mathrm{Gr}_{\leq\lambda} = \overline{\mathrm{Gr}_\lambda}$.

Proposition 3.3. *We have*

$$\dim \mathrm{Gr}_\lambda = \langle 2\rho, \lambda \rangle$$

where ρ is the half-sum of positive roots.

For dominant coweights λ, λ', we say $\lambda' \leq \lambda$ if $\lambda - \lambda'$ is a sum of positive coroots. The following proposition justifies the notation $\mathrm{Gr}_{\leq\lambda}$

Proposition 3.4. *Let λ be a dominant coweight. Then*

$$\mathrm{Gr}_{\leq\lambda} = \coprod_{\lambda' \in X_*(T)^+, \lambda' \leq \lambda} \mathrm{Gr}_{\lambda'}$$

Proof. We first show the containment:

$$\mathrm{Gr}_{\leq\lambda} \supset \bigcup_{\lambda' \leq \lambda, \lambda' \in X_*(T)^+} \mathrm{Gr}_{\lambda'}$$

It suffices to show that if $\lambda' = \lambda - \alpha^\vee$ for $\alpha^\vee$ a positive corrot, then

$$\mathrm{Gr}_{\lambda'} \subset \overline{\mathrm{Gr}_\lambda}$$

It suffices to show $t^{\lambda'} \in \overline{\mathrm{Gr}_\lambda}$. Let U_α be the subgroup corresponding to α and consider $U_\alpha(xt^{-1})$, a SL_2-computation shows that

$$\lim_{x \in \infty} U_\alpha(xt^{-1})t^\lambda = t^{\lambda'}$$

Conversely, we need to show that if $\mathrm{Gr}_{\lambda'} \subset \overline{\mathrm{Gr}_\lambda}$, then $\lambda' \leq \lambda$. Indeed, it's not hard to check that if $g(t) \in G(\mathcal{O})t^\lambda G(\mathcal{O})$ with λ dominant, and V is a finite dimensional G-module with lowest weight μ, then:

$$g(t) \cdot (V \otimes \mathcal{O}) \subset t^{\langle \lambda, \mu \rangle} V \otimes \mathcal{O}$$

This is a closed condition, in particular:

$$t^{\lambda'} \in \overline{\mathrm{Gr}_\lambda} \subset \{g(t) \mid g(t) \cdot (V \otimes \mathcal{O}) \subset t^{\langle \lambda, \mu \rangle} V \otimes \mathcal{O}\}$$

So $t^{\lambda'} \cdot (V \otimes \mathcal{O}) \subset t^{\langle \lambda, \mu \rangle} V \otimes \mathcal{O}$. By testing on $V = V_\mu$ for all μ, we see $\langle \lambda' - \lambda, \mu \rangle \geq 0$ for all μ. So $\lambda - \lambda'$ is a $\mathbb{Q}_{\geq 0}$-linear combination of positive roots.

To complete the proof, we need to argue it is actually a $\mathbb{Z}_{\geq 0}$-linear combination. This can be done by considering:

$$\mathrm{Gr}_G \to \pi_0(\mathrm{Gr}_G) = X_*(T)/\mathbb{Z}\Phi^\vee$$

This will send $t^\lambda \mapsto [\lambda]$. Therefore $t^\lambda, t^{\lambda'}$ are in the same component of Gr_G iff $\lambda - \lambda'$ are in the coroot lattice. We are therefore done. □

Now let's discuss singularities of $\mathrm{Gr}_{\leq\lambda}$. The slogan is that "the singularities of $\mathrm{Gr}_{\leq\lambda}$ is 50 percent of the content of the geometric Satake equivalence".

Example. *Let $G = \mathrm{GL}_n$, and $\mathcal{N}$ be the nilpotent cone of GL_n. Let $\mathrm{Gr}^{(n)}_{\mathrm{GL}_n}$ be the component of $\mathrm{Gr}_{\mathrm{GL}_n}$ consisting of lattices Λ in relative dimension n with the standard lattice $\mathcal{O}^n$. Lusztig showed that there is a G-equivariant open embedding:*

$$\mathcal{N} \hookrightarrow \mathrm{Gr}^+_{\mathrm{GL}_n} \cap \mathrm{Gr}^{(n)}_{\mathrm{GL}_n} = \mathrm{Gr}_{\leq(n,0,0,\cdots,0)}$$

inducing a bjection:

$$\mathrm{GL}_n \setminus \mathcal{N} \leftrightarrow \mathrm{GL}_n(\mathcal{O}) \setminus \mathrm{Gr}_{\leq(n,0,0,\cdots,0)}$$

The recipe sends $e \in \mathcal{N}$ to $\Lambda = (t+e)\mathcal{O}^n$. To check this is indeed an open embedding, we need to compare dimensions

$$\dim \mathrm{Gr}_{\leq(n,0,0\cdots,0)} = \langle 2\rho, (n,0,0,\cdots,0)\rangle = n(n-1) = \dim \mathcal{N}$$

Theorem 3.5. *(Lusztig) Suppose $\lambda' \leq \lambda$ are dominant coweights and $i_{\lambda'} : \mathrm{Gr}_{\lambda'} \hookrightarrow \mathrm{Gr}_{\leq\lambda}$. Then $i^*_{\lambda'}\mathrm{IC}(\mathrm{Gr}_{\leq\lambda})$ has total dimension $\dim V_\lambda(\lambda')$ concentrated in one parity, where V_λ is the irreducible representation of the dual group $G^\vee$ with highest weight λ, and $V_\lambda(\lambda')$ is its λ'-weight space.*

Remark. *This theorem is similar to how Kazhdan-Lusztig theory relates intersection cohomology sheaves of Schubert cells to the representation theory of $\mathfrak{g}$. This theorem is also the first time the Langlands dual group $G^\vee$ appears in geometric representation theory.*

4 Satake Isomorphism and Iwahori Hecke Algebra

Let's recall some classical theory or representation of p-adic groups. Let F be pa nonarchimedean local field of residue field $\mathbb{F}_q$. Let G/F be a split reductive group and $G = G(F)$, $K = G(\mathcal{O}_F)$. Recall that an irreducible representation π of $G(F)$ is spherical iff $\pi^K \neq 0$. Recall that:

Proposition 4.1. *All spherical representations appear in the spherical principal series, namely those of form:*

$$\mathrm{Ind}^G_B(\chi)$$

where χ is trivial on $T(\mathcal{O}_F)$, and therefore χ induces a character:

$$\chi : T(F)/T(\mathcal{O}_F) \simeq X_*(T) \to \mathbb{C}^\times$$

Therefore all such characters are parametrized by $T^\vee(\mathbb{C})$. Moreover, we have an extra symmetry given by W. So irreducible spherical representations of G are roughly classified by $T^\vee(\mathbb{C})/W$.

Another perspective is using spherical Hecke algebra:

$$\mathcal{H}_G^{\mathrm{sph}} = \mathbb{C}[K \setminus G/K]$$

Then $\pi \mapsto \pi^K$ induces an equivalence of categories between irreducible spherical represntations of G and simple $\mathcal{H}_G^{\mathrm{sph}}$-modules. It is a classical fact that $\mathcal{H}_G^{\mathrm{sph}}$ is commutative.

Remark. *The classical proof of commutativity of spherical Hecke algebra is using Gelfand trick. But we can also see this (in the case of* GL_n*) using Hall algebra. Let* $G = \mathrm{GL}_n(F)$*, and* $G^+ = \mathrm{GL}_n(F) \cap \mathrm{Mat}_n(\mathcal{O}_F)$*. Then* $\mathrm{Gr}^+_{\mathrm{GL}_n} := G^+/K$ *classifies* $\mathcal{O}_F$*-lattices* $\Lambda \subset \mathcal{O}_F^{\oplus n}$*.*

Now let X *be an algebraic curve with a point* $x_0 \in X(\mathbb{F}_q)$*. Let* $\mathrm{Tor}(X)_{x_0}$ *be the groupoid of torsion coherent sheaves on* X *supported at* x_0*. By choosing an isomorphism* $\hat{\mathcal{O}}_{X,x_0} \simeq \mathcal{O}_F$*, we get a map:*

$$\mathrm{Gr}^+_{\mathrm{GL}_n} \to \mathrm{Tor}(X)_{x_0}$$
$$\Lambda \mapsto \mathcal{O}_F^{\oplus n}/\Lambda$$

viewed as a torsion sheaf supported at x_0*. It's easy to see this descends to a map of groupoids:*

$$K \setminus G^+/K \to \mathrm{Tor}(X)_{x_0}$$

This induces a pullback map on finitely supported functions:

$$r : \mathbb{C}[\mathrm{Tor}(X)_{x_0}] \to \mathbb{C}[K \setminus G^+/K] =: \mathcal{H}_G^{\mathrm{sph},+}$$

The double cosets $K \setminus G^+/K$ *are in bijection with* $(\lambda_1 \geq \cdots \geq \lambda_n \geq 0)$*, since any* $g \in G^+$ *can be reduced to Smith normal form* $\mathrm{diag}(t^{\lambda_1}, \cdots, t^{\lambda_n})$ *with* $\lambda_i \geq 0$*. On the other hand, torsion sheaves on* X *supported on* x_0 *are classified by all partitions* $\lambda_1 \geq \cdots \geq 0$*. This shows* r *is surjective. Moreover, one can show that* r *is a ring homomorphism. Thus* $\mathcal{H}_G^{\mathrm{sph},+}$ *is commutative since Hall algebra is. The whole algebra* $\mathcal{H}_G^{\mathrm{sph}}$ *is obtained by* $\mathcal{H}_G^{\mathrm{sph},+}$ *via localization, therefore it's also commutative.*

Given an unramified character $\chi : T(F)/T(\mathcal{O}_F) \to \mathbb{C}^\times$, we can concretely write:

$$\mathrm{Ind}_B^G(\chi) = \{f : NT(\mathcal{O}) \setminus G \to \mathbb{C} \mid f \text{ is } \chi\text{-eigenfunction under } T(F)/T(\mathcal{O}) \text{ action}\}$$

$$\mathrm{Ind}_B^G(\chi)^K = \{f : NT(\mathcal{O}) \setminus G/K \to \mathbb{C} \mid f \text{ is } \chi\text{-eigenfunction under } T(F)/T(\mathcal{O}) \text{ action}\}$$

The *universal unramified principal series* is:

$$\mathbb{C}[NT(\mathcal{O}) \setminus G/K]$$

which has a left-action by $X_*(T)$ and right action by $\mathcal{H}_G^{\mathrm{sph}}$.

Lemma 4.2. *As a left* $\mathbb{C}[X_*(T)]$*-module,* $M = \mathbb{C}[NT(\mathcal{O}) \setminus G/K]$ *is free of rank 1.*

Proof. This is just Iwasawa decomposition. □

Therefore, we obtain a ring homomorphism:

$$\mathrm{Sat}^{\mathrm{naive}} : \mathcal{H}_G^{\mathrm{sph}} \to \mathrm{End}_{X_*(T)}(M) = \mathbb{C}[X_*(T)]$$

If we can show $\mathrm{Sat}^{\mathrm{naive}}$ is injective, then we have anotehr proof of $\mathcal{H}_G^{\mathrm{sph}}$ is commutative.

However, $\mathrm{Sat}^{\mathrm{naive}}$ is not an isomorphism, and to promote it to an isomorphism we must consider extra Weyl group symmetries. To see this, we need to make our naive Satake map explicit. Since $\mathbf{1}_{NK}$ is a basis for M as a left $\mathbb{C}[X_*(T)]$-module, we have:

$$\mathrm{Sat}^{\mathrm{naive}} : \mathcal{H}_G^{\mathrm{sph}} \to \mathbb{C}[X_*(T)]$$

$$f \mapsto (\lambda \mapsto (\mathbf{1}_{NK} * f)(t^\lambda))$$

However, we need a normalization:

$$\mathrm{Sat} : \mathcal{H}_G^{\mathrm{sph}} \to \mathbb{C}[X_*(T)]$$

$$f \mapsto (\lambda \mapsto (q^{\langle \rho, \lambda \rangle} 1_{NK} * f)(t^\lambda))$$

With this normalization, we have:

Theorem 4.3 (Satake)**.** *The map* Sat *is an isomorphism of* $\mathbb{C}$*-algebras:*

$$\mathrm{Sat} : \mathcal{H}_G^{\mathrm{sph}} \simeq \mathbb{C}[X_*(T)]^W$$

Corollary 4.4. *Irreducible spherical representations of* G *are classified by* $T^\vee(\mathbb{C})/W$*. In fact, for* $\chi \in T^\vee(\mathbb{C})$*, the corresponding irreducible spherical representation of* G *appears as a subquotient of the normalized induction* $\mathrm{Ind}_B^G(\chi|-|^\rho)$*, which has a one-dimensional space of* K*-fixed vectors.*

Let's now briefly discuss Iwahori-Hecke algebras. Recall that finite Hecke algebra is defined to be $\mathbb{C}[B(\mathbb{F}_q)\backslash G(\mathbb{F}_q)/B(\mathbb{F}_q)]$, with generators T_s satisfying the relation $(T_s+1)(T_s-q)=0$ as well as the braid relations. Indeeed, this makes sense for any Coxeter group (W,S), and we can define an algebra $\mathcal{H}_q(W)$, which is a specialization of the generic Hecke algebra $\mathcal{H}_v(W)$.

Let $Q^\vee \subset X_*(T)$ be coroot lattice and cocharacter lattice. Then the affine Weyl group is defined to be:

$$W_a = W \ltimes Q^\vee$$

which is a subgroup of the extended affine Weyl group

$$\widetilde{W} = W \ltimes X_*(T)$$

If G is simple, then W_a is a Coxeter group with $S = S_a = S \cup \{s_0\}$, where s_0 is the reflection across a wall of the fundamental alcove. Explicitly, $s_0 = (\theta^\vee) \cdot s_\theta$.

Recall that the Iwahori subgroup of G is defined to be:

$$I \coloneqq \pi^{-1}(B(k))$$

for $\pi : G(\mathcal{O}_F) \to G(\mathbb{F}_q)$.

Proposition 4.5. *When G is simple and simply connected, there is a bijection:*

$$W_a \xrightarrow{\sim} I \setminus G/I$$
$$\lambda \in X_*(T) \mapsto t^\lambda$$
$$w \mapsto I\dot{w}I$$

For an element as written in the semidirect product, we just lift them individually and take product. For example:

$$s_0 \mapsto It^{\theta^\vee}\dot{s}_\theta I$$

Now $\mathcal{H}_q(W_a)$ has a representation-theoretic meaning. Let $I \subset G$ be the Iwahori, then:

Proposition 4.6. *Let G be simple, simply connected group, there is an isomorphism of algebras:*

$$\mathcal{H}_q(W_a) \to \mathbb{C}[I \setminus G/I]$$
$$T_w \mapsto \mathbf{1}_{I\dot{w}I}$$

Here $w \in W_a$. For example, if $s \in S$ is a finite reflection, then:

$$T_s \mapsto \mathbf{1}_{B(\mathbb{F}_q)\dot{s}B(\mathbb{F}_q)} \in \mathbb{C}[B(\mathbb{F}_q) \setminus G(\mathbb{F}_q)/B(\mathbb{F}_q)] = \mathbb{C}[I \setminus K/I] \subset \mathbb{C}[I \setminus G/I]$$

Kazhdan and Lusztig introduced Kazhdan-Lusztig polynomials for every Coxeter group, and in particular they exist for affine Weyl groups. Let W_a be an affine Weyl group, $\mathcal{H}_v(W_a)$ be a generic affine Hecke algebra. It has standard basis $\{T_w\}$ and Kazhdan Lusztig basis $\{C'_w\}$. Then the Kazhdan-Lusztig polynomial is the transition matrix for these two basis:

$$C'_w = v^{-\ell(w)} \sum_{y \le w} P_{y,w}(v^2) T_y$$

with $P_{w,w}(v^2) = 1$.

The spherical Hecke algebra is a subset of the Iwahori Hecke algebra:

$$i : \mathcal{H}_G^{\mathrm{sph}} = \mathbb{C}[K \setminus G/K] \hookrightarrow \mathbb{C}[I \setminus G/I] = \mathcal{H}_q(W_a)$$
$$\mathbf{1}_{Kt^\lambda K} \mapsto \sum_{w \in W \cdot t^\lambda W} T_w$$

This is **not** an algebra homomorphism: the unit $\mathbf{1}_K$ is not sent to unit $\mathbf{1}_I$.

Fact: The Kazhdan-Lusztig basis $C'_w \in \mathcal{H}_G^{\mathrm{sph}}$ iff w is longest in its W-coset WwW.

For general G, we replace W_a by extended affine Weyl group $\widetilde{W}$, and $\mathcal{H}_q(W_a)$ by $\mathcal{H}_q(\widetilde{W})$. Then:

$$\mathbb{C}[I \setminus G/I] \simeq \mathcal{H}_q(\widetilde{W})$$

as $\mathbb{C}$-algebras.

Now there is a bijection:

$$X_*(T)^+ \to W \setminus \widetilde{W}/W$$
$$\lambda \mapsto Wt^\lambda W$$

Let d_λ be the longest element in the coset $Wt^\lambda W$. One can show that:

$$\ell(d_\lambda) = \ell(w_0) + \langle 2\rho, \lambda \rangle$$

(the number $\langle 2\rho, \lambda \rangle$ appeared before as $\dim \mathrm{Gr}_\lambda$). By the fact above, we know $C'_{d_\lambda} \in \mathcal{H}_G^{\mathrm{sph}}$.

Theorem 4.7 (Satake, Lusztig, Casselman-Shalika)**.** *We have:*

1. *The elements* $\{C'_{d_\lambda}\}_{\lambda \in X_*(T)^+}$ *form a basis for* $\mathcal{H}_G^{\mathrm{sph}}$.
2. *We have* $\mathrm{Sat}(C'_{d_\lambda}) \in \mathbb{Z}[X_*(T)]^W$ *is independent of* q. *More precisely:*

$$\mathrm{Sat}(C'_{d_\lambda}) = \chi_{V_\lambda} = \sum_{\mu \in X_*(T)} \dim V_\lambda(\mu) 1_\mu$$

where V_λ *is the highest weight module of Langlands dual group* $G^\vee$ *with highest weight* λ .

Remark. *This makes sense since* $\lambda \in X_*(T)^+$ *can be viewed as an element in* $X^*(T^\vee)^+$, *which thus gives a highest weight module* V_λ *for the dual group* $G^\vee$.

5 Convolution Product

The goal of this section is to construct a convolution product on $D_{G(\mathcal{O})}(\mathrm{Gr})$: the Satake category of $G(\mathcal{O})$-equivariant sheaves on the affine Grassmanian. Note this categorifies the Hecke algebra $\mathcal{H}_G^{\mathrm{sph}}$, and we expect the convolution product makes $D_{G(\mathcal{O})}(\mathrm{Gr})$ a monoidal category.

As always, we work with $\overline{\mathbb{Q}_\ell}$-sheaves. First, let's define precisely what is $D_{G(\mathcal{O})}(\mathrm{Gr})$. Recall that:

$$\mathrm{Gr}_{\leq \lambda} \subset \mathrm{Gr} = \bigcup_{\lambda \in X_*(T)^+} \mathrm{Gr}_\lambda$$

is projective, we have a $G(\mathcal{O})$-action on $\mathrm{Gr}_{\leq \lambda}$. This action will factor through some finite-dimensional quotient $G(\mathcal{O})_N$ of $G(\mathcal{O})$. We thus define:

$$D_{G(\mathcal{O})}(\mathrm{Gr}_{\leq \lambda}) \coloneqq D_{G(\mathcal{O})_N}(\mathrm{Gr}_{\leq \lambda})$$

Note that for N sufficiently large the RHS is independent of N, because $\ker(G(\mathcal{O})_{N+1} \to G(\mathcal{O})_N)$ is unipotent that acts trivially on $\mathrm{Gr}_{\leq \lambda}$.

Let $\omega \in \pi_0(\mathrm{Gr}) = \pi_1(G) = X_*(T)/Q^\vee$. Then the connected component corresponding to ω can be written as:

$$\mathrm{Gr}^\omega = \bigcup_{\lambda, \overline{\lambda} = \omega} \mathrm{Gr}_{\leq \lambda}$$

and hence we can define:

$$D_{G(\mathcal{O})}(\mathrm{Gr}^\omega) = \bigcup_{\lambda, \overline{\lambda} = \omega} D_{G(\mathcal{O})}(\mathrm{Gr}_{\leq \lambda})$$

and also $D_{G(\mathcal{O})}(\mathrm{Gr}) = \bigcup_\omega D_{G(\mathcal{O})}(\mathrm{Gr}^\omega)$.

Now in order to define the convolution product, first look at what's happening on the function level. Consider the map:

$$G(F) \times^{G(\mathcal{O})} G(F)/G(\mathcal{O}) \to G(F)/G(\mathcal{O})$$

given by multiplication. This gives rise to:

$$\mathbb{C}[G(F)/G(\mathcal{O})] \times \mathcal{H}_G^{\mathrm{sph}} \to \mathbb{C}[G(F)/G(\mathcal{O})]$$

that acts by

$$(f \cdot h)(x) = \sum_{g \in G(F)/G(\mathcal{O})} f(g)h(g^{-1}x)$$

The goal is to geometrize this map. To this end, we replace $G(F)$ by the loop group LG:

$$LG(R) = G(R((t))) = \bigcup G(\text{“}t^{-n}R[[t]]\text{”})$$

The quotation means that they are matrices with entries in the specified form after choosing an embedding $G \hookrightarrow \mathrm{GL}_N$. Ecah of these spaces is a closed subset of $\mathbb{A}^\infty$, and so this functor is representated by an ind-scheme.

Additionally, $G(\mathcal{O})$ is replaced by the positive loop group L^+G. This is a functor that sends k-algebra R to $G(R[[t]])$. This functor is represented by a scheme. The quotient $G(F)/G(\mathcal{O})$ is replaced by affine Grassmannian Gr. The map from above now becomes a convolution map:

$$LG \times^{L^+G} \mathrm{Gr} \xrightarrow{m} \mathrm{Gr}$$

We also have the projection map:

$$p_1 : LG \times^{L^+G} \mathrm{Gr} \to LG/L^+G = \mathrm{Gr}$$

This induces an isomorphism:

$$LG \times^{L^+G} \mathrm{Gr} \xrightarrow{(p_1,m)} \mathrm{Gr} \times \mathrm{Gr}$$

Now let $LG_{\leq\lambda}$ be the preimage of $\mathrm{Gr}_{\leq\lambda}$ in LG. Let us denote $\mathrm{Gr}^{(2)}_{\leq(\lambda,\mu)} \coloneqq LG_{\leq\lambda} \times^{L^+G} \mathrm{Gr}_{\leq\mu}$.

Example. *Consider the case $G = \mathrm{GL}_n$. Recall that $\mathrm{Gr}(F) = \{\mathcal{O}_F$-lattices $\Lambda \subset F^n\}$, and $\mathrm{Gr} \times \mathrm{Gr}(F) = \{\Lambda_1, \Lambda_2 \subset F^n\}$. It follows that:*

$$\mathrm{Gr}^{(2)}_{\leq(\lambda,\mu)} = \{(\Lambda_1, \Lambda_2) \mid \mathrm{pos}(\mathcal{O}^n, \Lambda_1) \leq \lambda, \mathrm{pos}(\Lambda_1, \Lambda_2) \leq \mu\}$$

is a projective variety. The multiplication thus gives:

$$m_{\lambda,\mu} : \mathrm{Gr}_{\leq(\lambda,\mu)} \to \mathrm{Gr}_{\leq\lambda+\mu}$$

$$(\Lambda_1, \Lambda_2) \mapsto \Lambda_2$$

In fact we will see this is a surjection.

Now we can draw the convolution diagram:

$$\begin{array}{ccccc} & & LG_{\leq\lambda} \times^{L^+G} \mathrm{Gr}_{\leq\mu} & \xrightarrow{\quad m_{\lambda,\mu} \quad} & \mathrm{Gr}_{\leq\lambda+\mu} \\ & \swarrow^{p_1} & & \searrow^{p_2} & \\ \mathrm{Gr}_{\leq\lambda} & & & & L^+G\backslash\mathrm{Gr}_{\leq\mu} \end{array}$$

Now let $\mathcal{F}, \mathcal{G} \in D_{G(\mathcal{O})}(\mathrm{Gr})$ with supp $(\mathcal{F}) \subset \mathrm{Gr}_{\leq\lambda}$, supp $(\mathcal{G}) \subset \mathrm{Gr}_{\leq\mu}$. Note that $\mathcal{G} \in D(L^+G \backslash \mathrm{Gr}_{\leq\mu})$ due to L^+G-equivariance of $\mathcal{G}$. We can thus define the convolution product:

$$\mathcal{F} * \mathcal{G} = m_{\lambda,\mu,*}(p_1^*\mathcal{F} \otimes p_2^*\mathcal{G})$$

This is supported on $\mathrm{Gr}_{\leq\lambda+\mu}$. Direct computation shows:

Proposition 5.1. *$(D_{G(\mathcal{O})}(\mathrm{Gr}), *)$ is a monoidal category with unit given by skyscraper δ at $1 \in \mathrm{Gr}$.*

6 Perverse Sheaves on Affine Grassmannian

Now let $P_{G(\mathcal{O})}(\mathrm{Gr})$ be the abelian category of $G(\mathcal{O})$-equivariant perverse sheaves on Gr. Then:

Theorem 6.1. *The category $P_{G(\mathcal{O})}(\mathrm{Gr})$ is closed under $*$, which equips $P_{G(\mathcal{O})}(\mathrm{Gr})$ with the structure of a monoidal category.*

Remark. *The analogous statement in the setting of finite or affine Hecke category is false.*

In fact, moreover:

Theorem 6.2. *$P_{G(\mathcal{O})}(\mathrm{Gr})$ has a commutativity constraint, making it into a tensor category.*

Note that the simple perverse sheaves are:

$$\mathrm{IC}_\lambda = \mathrm{IC}(\mathrm{Gr}_{\leq\lambda})$$

It turns out that there are no nontrivial $G(\mathcal{O})$-equivariant local systems on Gr_λ due to connective stablizers. Another way to see this:

$$\mathrm{Gr}_\lambda = G(\mathcal{O})/G(\mathcal{O}) \cap t^\lambda G(\mathcal{O}) t^{-\lambda}$$

This sits inside the affine bundle:

$$\mathrm{Gr}_\lambda \to G/P_{-\lambda}$$

where $\mathrm{Lie}P_{-\lambda} = \mathfrak{t} \oplus (\oplus_{\langle\lambda,\alpha\rangle\leq 0}\mathfrak{g}_\alpha)$. This proves that Gr_λ is simply connected, hence there are no nontrivial local systems on Gr_λ. From this description we also see that $\dim \mathrm{Gr}_\lambda = \langle 2\rho, \lambda\rangle$. In particular:

Corollary 6.3. *On a fixed component of* Gr*, all $G(\mathcal{O})$-orbits dimensions have same parity.*

This follows by noting that if $\mathrm{Gr}_\lambda, \mathrm{Gr}_\mu$ are in the same component, then $\lambda - \mu \in X_*(T)$.

Example. *Let $G = \mathrm{GL}_n$. Consider connected components of* Gr *in this case, which are labelled by an integer $[\Lambda : \mathcal{O}^n] = -m$ for some integer m. The smallest Schubert variety in the $m = 0$ component is a single point $\mathcal{O}^n$. The smallest one in the $m = 1$ component is given by lattices $\Lambda \subset \mathcal{O}^n$ with relative dimension 1: this is $\mathbb{P}^{n-1}$. Continuing this was it gives that the smallest component is $\mathrm{Gr}\binom{n}{m}$ for $m < n$. There are the n-periodic with the map from the m-component to the $m+n$ component given by multiplication by t. Note that if n is even, then the dimension of these Schubert varieties alternate in parity; otherwise everything is even.*

We now know that $P_{G(\mathcal{O})}(\mathrm{Gr})$ has simple objects given by $\{\mathrm{IC}_\lambda\}_{\lambda \in X_*(T)^+}$. Recall that $\mathcal{H}_G^{\mathrm{sph}}$ contains the KL basis C'_{d_λ}, where d_λ is the longest element in $W \cdot \lambda \cdot W \subset \widetilde{W}$. Now let

$$\mathrm{ev}_0 : G(\mathcal{O}) \to G(k)$$

be the map $t \mapsto 0$, and we define the *Iwahori* to be $\mathrm{ev}_0^{-1}(B(k)) \subset G(\mathcal{O})$. The *affine flag variety* is then defined to be:

$$\mathrm{Fl} \coloneqq G(F)/I$$

Then the categorification of the affine Hecke algebra is given by Iwahori equivariant sheaves $D_I(\mathrm{Fl})$ on the affine flag variety. This category contains IC_w for $w \in \widetilde{W}$. These IC-sheaves correspond to KL basis C'_w. Now that for d_λ:

$$\mathrm{IC}_{d_\lambda} = \mathrm{IC}(\overline{Id_\lambda I/I})$$

We have the Cartesian diagram:

$$\begin{array}{ccc} \overline{Id_\lambda I/I} = \pi^{-1}(\mathrm{Gr}_{\leq \lambda}) & \hookrightarrow & \mathrm{Fl} \\ \downarrow & & \downarrow \pi \\ \mathrm{Gr}_{\leq \lambda} & \hookrightarrow & \mathrm{Gr} \end{array}$$

By smooth pullback of perverse sheaves:

$$\mathrm{IC}_{d_\lambda} = \pi^* \mathrm{IC}_\lambda[\dim \pi]$$

Theorem 6.4. *The category $P_{G(\mathcal{O})}(\mathrm{Gr})$ is semisimple.*

Proof. It suffices to show that $\mathrm{Ext}^1(\mathrm{IC}_\lambda, \mathrm{IC}_\mu) = 0$, where λ, μ have the same image in $\pi_1(G)$. The idea is to use parity. Let's assume both Gr_λ and Gr_μ are even-dimensional, so that $\mathrm{IC}_\lambda = j_{!*}\overline{\mathbb{Q}_\ell}[\mathrm{even}]$. Recall in Hecke category, the stalks and costalks of IC sheaves have parity vanishing. The same argument can be adapted here for the affine Hecke category as well. It follows that $\mathrm{IC}_\lambda, \mathrm{IC}_\mu$ have stalks and costalks live in even degrees. Now $\mathrm{Ext}^*(\mathrm{IC}_\lambda, \mathrm{IC}_\mu)$ can be computed from a spectral sequence via terms of the form

$$E_1 = i_\nu^! \mathrm{Ext}^*(\mathrm{IC}_\lambda, \mathrm{IC}_\mu) = \mathrm{Ext}^*(i_\nu^* \mathrm{IC}_\lambda, i_\nu^! \mathrm{IC}_\mu)$$

,where $i_\nu : \mathrm{Gr}_\nu \hookrightarrow \mathrm{Gr}$: this is a general property for hypercohomology for filtered topological spaces. Here ν has the same image in $\pi_1(G)$ as λ, μ. These terms are given by sums of $H^*(G/P_{-\nu})$ shifted by even degrees. This then shows that odd Ext vanishing, and in particular Ext^1. □

Let's now discuss the idea of proving Theorem 6.1. We wish to show that $\mathrm{IC}_\lambda * \mathrm{IC}_\mu$ is perverse. Namely we wish to calculate the stalks on each strata Gr_ν. In fact, it suffices to compute the stalk at t^ν. Recall the convolution product is defined via the convolution diagram:

$$\begin{array}{ccccc} \mathrm{Gr}^{(2)}_{\leq(\lambda,\mu)} = LG_{\leq\lambda} \times^{L^+G} \mathrm{Gr}_{\leq\mu} & & \xrightarrow{\quad m_{\lambda,\mu} \quad} & & \mathrm{Gr}_{\leq\lambda+\mu} \\ \swarrow^{p_1} & & & \searrow^{p_2} & \\ \mathrm{Gr}_{\leq\lambda} & & & & L^+G\backslash\mathrm{Gr}_{\leq\mu} \end{array}$$

Therefore:

$$\mathrm{IC}_\lambda * \mathrm{IC}_\mu = m_{\lambda,\mu,*}\mathrm{IC}_{\lambda,\mu}$$

So by proper base change:

$$(\mathrm{IC}_\lambda * \mathrm{IC}_\mu)|_{t^\nu} = H^*_c(m^{-1}_{\lambda,\mu}(t^\nu), \mathrm{IC}_{\lambda,\mu})$$

Recall that $\dim \mathrm{Gr}_\nu = \langle 2\rho, \nu\rangle$, then the (upper) perverse condition asks us to show that the degree of the above cohomology concentrate in $\leq -\langle 2\rho, \nu\rangle$. Now we have stratification:

$$\mathrm{Gr}^{(2)}_{\leq(\lambda,\mu)} = \coprod_{\lambda'\leq\lambda,\mu'\leq\mu} \mathrm{Gr}^{(2)}_{\lambda',\mu'}$$

Then intersect fibres with strata, we get:

$$H^*_c(m^{-1}_{\lambda,\mu}(t^\nu) \cap \mathrm{Gr}^{(2)}_{\lambda',\mu'}, \mathrm{IC}_{\lambda,\mu}\mid_{\mathrm{Gr}^{(2)}_{\lambda',\mu'}})$$

Note that $\mathrm{IC}_{\lambda,\mu}$ is essentially $\mathrm{IC}_\lambda \boxtimes \mathrm{IC}_\mu$, so it concentrates in degree $\leq -\langle 2\rho, \lambda'\rangle - \langle 2\rho, \mu'\rangle$. In fact we saw that there can be no nontrivial extensions between equivariant perverse sheaves (using a parity argument), so the restriction is actually a direct sum of shifted constant sheaves.

Recall that the goal is to show that the degree $\leq -\langle 2\rho, \nu\rangle$, so it suffices to show:

$$\dim(m^{-1}_{\lambda,\mu}(t^\nu) \cap \mathrm{Gr}^{(2)}_{\lambda',\mu'}) \leq \langle \rho, \lambda' + \mu' - \nu\rangle$$

But note that $m^{-1}_{\lambda,\mu}(t^\nu) \cap \mathrm{Gr}^{(2)}_{\lambda',\mu'} \subset m_{\lambda',\mu'}(t^\nu)$. So it suffices to show:

$$\dim(m^{-1}_{\lambda,\mu}(t^\nu) \cap \mathrm{Gr}^{(2)}_{\lambda,\mu}) \leq \langle \rho, \lambda + \mu - \nu\rangle \tag{1}$$

Remark. *This is an analogue statement for:*

$$\dim \mathcal{B}_e \leq \frac{1}{2}(\dim \mathcal{N} - \dim \mathcal{O}_e)$$

Let's quickly remark how to formally check the perversity condition on the Verdier dual. This is because the whole convolution construction commutes with Verdier dual. Indeed, we construct the convolution product via "pull-push". But the pull process is essentially a "product" situation (at least etale locally), and we pushforward along a proper map. Therefore $\mathbb{D}(\mathcal{F} * \mathcal{G}) = \mathbb{D}\mathcal{F} * \mathbb{D}\mathcal{G}$. Hence the other perversity condition follows.

Hence it remains to check the geometric statement (1). This will need more understanding of the geometry of affine Grassmannian.

7 Semi-Infinite Orbits

We now want to construct a geometric version of unramified principal series. Recall that the universal unramified principal series is:

$$\mathbb{C}[N(F) \setminus G(F)/G(\mathcal{O})]$$

Note that the loop group LN acts on Gr in the obvious way, and the Iwasawa decomposition asserts that the orbits are in bijection with $X_*(T)$. These orbits are both infinite dimensional and have infinite codimension: these are called *semi-infinite orbits*. Let S_λ be the LN orbit of t^λ. It will have a similar closure relation to to the Schubert varieties, namely: $\overline{S_\lambda} = \bigcup_{\mu \leq \lambda} S_\mu$. Here μ runs over all coweights such that $\lambda - \mu$ is a sum of positive coroots. Similarly, letting N^- be the opposite unipotent subgroup of G, and let T_λ denote the LN^- orbit of t^λ.

Remark. *In fact, more is true. $\overline{S_\lambda} \setminus S_\lambda$ is a Cartier divisor of $\overline{S_\lambda}$, cutting out by the vanishing of one Plücker coordinate with respect to the embedding:*

$$\overline{S_\lambda} \subset \mathrm{Gr}_G \subset \mathrm{Gr}_N \binom{2Nn}{Nn}$$

where the second inclusion is for $G = \mathrm{SL}_n$.

Consider $T_c \subset T(\mathbb{C})$ acts on $\mathrm{Gr}(\mathbb{C})$, it turns out that this is a Hamiltonian action, hence we have a moment map:

$$\mu : \mathrm{Gr}(\mathbb{C}) \to (\mathrm{Lie} T_c)^*$$

The different subsets of affine Grassmannian introduced so far can be visualized by taking their images under this moment map. For a closed subset of $Z \subset \mathrm{Gr}(\mathbb{C})$, $\mu(Z)$ will be the convex hall of $\mu(t^\lambda)$ such that $t^\lambda \in Z$. In this perspective, $\mu(\mathrm{Gr}_{\leq \lambda})$ is the convex hall of $W \cdot \lambda$ and $\mu(\overline{S_\lambda})$ is the convex hall of $\lambda - \mathrm{Span}_{\mathbb{R}_{\geq 0}}(\alpha^\vee \mid \alpha > 0)$.

By looking at these types of pictures, the extreme cases of the intersections of $S_{\lambda'}$ and $\mathrm{Gr}_{\leq \lambda}$ can be seen. Namely, when $\lambda' = w_0 \lambda$ we have $S_{w_0 \lambda} \cap \mathrm{Gr}_{\leq \lambda} = \{t^{w_0 \lambda}\}$; when $\lambda' = \lambda$, we have $S_\lambda \cap \mathrm{Gr}_{\leq \lambda}$ is dense in $\mathrm{Gr}_{\leq \lambda}$. In fact, we have the following more general statement.

Theorem 7.1. *1. $S_\mu \cap \mathrm{Gr}_{\leq \lambda} \neq \emptyset$ iff $\mu \in \mathrm{Conv}(W \cdot \lambda)$.*

2. When non-empty, $S_\mu \cap \mathrm{Gr}_{\leq \lambda}$ is equidimensional with dimneions $\langle \rho, \lambda + \mu \rangle$.

Remark. *There are a lot of things not justified. For example, we must show that for a vertex v in a moment polytope, $\mu^{-1}(x) = \{t^\lambda\}$ for some λ is a single point. Let's sketch a proof of this fact. First one sees that $\{t^\lambda\}$ are precisely the fixed points for the T-action. Let x be not fixed by T such that $\mu(x)$ is a vertex. Then we can pick a generic $\mathbb{G}_m \subset T$ such that x is not fixed by $\mathbb{G}_m$. This $\mathbb{G}_m$-action will lead to two limit points y, z distinct from x, and they will be fixed by T (by choosing $\mathbb{G}_m \subset T$ generic enough, and use the fact that $\mathrm{Gr}_{\leq \lambda}$ is a projective scheme). Now that consider the moment map of this $\mathbb{G}_m$-action, we get:*

$$\mu' : \mathrm{Gr}_{\leq \lambda} \to \mathbb{R}$$

One sees that $\mu'(x)$ sits strictly inside the interval with endpoints $\mu'(y), \mu'(z)$. This contradicts the fact that $\mu(x)$ is an endpoint.

Definition 7.2. The irreducible components of such intersections are called MV-cycles.

It turns out that MV-cycles in $S_\mu \cap \mathrm{Gr}_{\leq \lambda}$ will give a basis of $V_\lambda(\mu)$.

Example. *Let $G = \mathrm{GL}_n$, then our semi-infinite orbit:*

$$S_\lambda \subset \mathrm{Gr} = \{\Lambda \subset F^n\}$$

What conditions cut out S_λ? Note that $G(F)/N(F)$ is the moduli of "framed flags" over F: a choice of flag $V_1 \subset V_2 \subset \cdots \subset V_n$ together with $e_i \in \mathrm{gr}_i V_\bullet$. We fix the standard flag with standard framing:

$$0 \subset \langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \cdots \subset \langle e_1, \cdots, e_n \rangle$$

with $e_i \in \mathrm{gr}_i$. Now observe that

$$G(F) \setminus (G(F)/N(F) \times G(F)/G(\mathcal{O})) \simeq N(F) \setminus G(F)/G(\mathcal{O}) \simeq X_*(T)$$

Given a lattice Λ, we can consider:

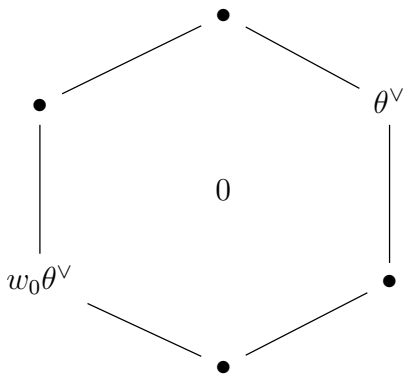
$$\Lambda_i = \Lambda \cap \langle e_1, \cdots, e_i \rangle \subset F^i$$

this is a lattice. Then $\Lambda_i/\Lambda_{i-1} \subset F^i/F^{i-1} = \langle e_i \rangle$ is rank one lattice. We define $\lambda_i \in \mathbb{Z}$ be such that:

$$\Lambda_i/\Lambda_{i-1} = \mathcal{O} \cdot t^{\lambda_i} e_i$$

This cut out the semi-infinite orbit S_λ. For example: S_0 consists of all lattices such that $\Lambda_i/\Lambda_{i-1} = \mathcal{O} \cdot e_i$ for all i. Note that on the closure of $\overline{S}_\lambda$, the intersection only gets bigger (just like for Schubert cells). This aligns with our result stating that the closure of $\overline{S}_\lambda$ will contain orbits corresponding to smaller cocharacters.

Example. *Let's consider $G = \mathrm{SL}_3$.*



where $\theta^\vee = (1, 0, -1)$ is the coroot corresponding to the highest root. In general we have a map:

$$\mathcal{O}_{\min} \hookrightarrow \mathrm{Gr}_{\theta^\vee}$$
$$g \cdot X_\theta \mapsto 1 + t^{-1} g \cdot X_\theta$$

where $\mathcal{O}_{\min}$ is the minimal nilpotent orbit, $0 \neq X_\theta \in \mathfrak{g}_\theta$ an element in the highest root space, and $\mathcal{O}_{\min} = \mathrm{Ad}G \cdot X_\theta$. This map is well-defined since:

$$1 + t^{-1} g \cdot X_\theta = g \cdot (1 + t^{-1} X_\theta)$$

but then $1 + t^{-1} X_\theta$ is a loop group element because it lies in the image of the embedding:

$$\mathrm{SL}_2 \xrightarrow{\phi_\theta} G$$

In fact, this is an open embedding and misses precisely those fixed points under loop rotation. It turns out that for general G, we have:

$$\overline{S_0} \cap \mathrm{Gr}_{\theta^\vee} \supset \overline{S_0} \cap \mathcal{O}_{\min} = \mathfrak{n} \cap \mathcal{O}_{\min}$$

Recall that in Springer theory, we know:

$$G \times^B (\mathfrak{n} \cap \mathcal{O}_e) = \pi^{-1}(\mathcal{O}_e)$$

Therefore the irreducible components of $\mathfrak{n} \cap \mathcal{O}_e$ biject with $\mathrm{Irr}(\mathcal{B}_e)/A_e$. Under Springer correspondence, the action of W on $\mathbb{C}[\mathrm{Irr}(\mathfrak{n} \cap \mathcal{O}_e)]$ is the representation attached to $(\mathcal{O}_e, \mathsf{triv})$. For us $\mathcal{O}_e = \mathcal{O}_{\min}$, we have:

$$(\mathcal{O}_{\min}, \mathsf{triv}) \xleftrightarrow{-\otimes \mathsf{sgn}} (\mathcal{O}_{\mathrm{subreg}}, \mathsf{triv})$$

But recall that $(\mathcal{O}_{\mathrm{subreg}}, \mathsf{triv})$ is the reflection representation, so that $(\mathcal{O}_{\min}, \mathsf{triv})$ is reflection tensor with sign. In particular, this tells us:

$$|\mathrm{Irr}(\mathfrak{n} \cap \mathcal{O}_{\min})| = \mathrm{rk}\ G_{\mathrm{Ad}}$$

So the number of components of $\overline{S_0} \cap \mathrm{Gr}_{\leq \theta^\vee}$ is $\mathrm{rk}\ G_{\mathrm{Ad}}$.

Proof. (dimension for MV cycles, sketched). We check extreme case $\mu = w_0 \lambda$ by brute force. In general, we take a path $\lambda \to w_0 \lambda$ going through μ. We show that at each step (substracting a simple coroot), the dimension drop by 1. □

Recall in last section, we promised to prove the bound (1):

$$\dim(m_{\lambda,\mu}^{-1}(t^\nu) \cap \mathrm{Gr}_{\lambda,\mu}^{(2)}) \leq \langle \rho, \lambda + \mu - \nu \rangle$$

for λ, μ, ν all dominant.

Proof. It suffices to show for ν anti dominant, we have:

$$\dim m_{\lambda,\mu}^{-1}(t^\nu) \cap \mathrm{Gr}_{\leq(\lambda,\mu)}^{(2)} \leq \langle \rho, \mu + \lambda + \nu \rangle$$

Let's just work in the case of GL_n. Note that:

$$Z = m_{\leq(\lambda,\mu)}^{-1}(t^\nu) \cap \mathrm{Gr}_{\lambda,\mu}^{(2)} = \{\Lambda \mid \Lambda_0 \xrightarrow{\leq \lambda} \Lambda \xrightarrow{\leq \mu} t^\nu \Lambda_0\}$$

Therefore we obtain a closed embedding:

$$Z \hookrightarrow \mathrm{Gr}_{\leq\lambda},$$

Now consider $T = T(k)$ acts on Gr by left multiplication. Since $\Lambda_0, t^\nu\Lambda_0$ are both fixed by T-action, we know Z is T-stable. But then what are the fixed points? In fact, it's not hard to see:

$$Z^T = \{\nu' \mid \Lambda_0 \xrightarrow{\leq\lambda} t^{\nu'}\Lambda_0 \xrightarrow{\leq\mu} t^\nu\Lambda_0\}$$

This is equivalent to ask for:

$$\nu' \in \mathrm{Conv}(W\cdot\lambda), \quad \nu - \nu' \in \mathrm{Conv}(W\cdot\mu)$$

Now generality of torus action tells me:

$$Z = \coprod_{\nu': t^{\nu'}\in Z^T} (S_{\nu'} \cap Z)$$

because $S_{\nu'}$ are the attracting cells for a regular dominant $\mathbb{G}_m$ action on Gr for the point $t^{\nu'}$. Therefore:

$$\dim Z = \max_{\nu': t^{\nu'}\in Z^T} \dim(S_{\nu'}\cap Z) \leq \max_{\nu'} \dim(S_{\nu'}\cap \mathrm{Gr}_{\leq\lambda}) = \max_{\nu'} \langle\rho, \lambda+\nu'\rangle$$

Hence we only need to show if ν' satisfies:

$$\nu' \in \mathrm{Conv}(W\cdot\lambda), \quad \nu - \nu' \in \mathrm{Conv}(W\cdot\mu)$$

then $\langle\rho, \mu + (\nu-\nu')\rangle \geq 0$. But this is obvious. □

8 Geometric Satake Equivalence

We already proved that:

$$(P_{L^+G}(\mathrm{Gr}), *)$$

is a monoidal category. In this section, we will give it extra structure so that:

$$P_{L^+G}(\mathrm{Gr}) \simeq \mathrm{Rep}(G^\vee)$$

as *tensor categories* (in particular both sides will have compatible isomorphisms $V\otimes W \simeq W\otimes V$). In addition, both sides are rigid in the sense that there are *dual objects*. For the LHS, the dual of $\mathcal{F}$ will be:

$$\mathbb{D}(\mathrm{inv}^*\mathcal{F})$$

where $\mathrm{inv} : LG \to LG$ (think of $\mathcal{F} \in D(L^+G \setminus LG/L^+G)$ as a sheaf on LG). This is **Geometric Satake Equivalence**. Notably this gives a geometric construction of $G^\vee$ because Tannakian reconstruction theorem allows us to reconstruct $G^\vee$ from the tensor category $(\mathrm{Rep}(G^\vee), \otimes)$.

Note that both sides have fibre functors:

$$\mathrm{Rep}(G^\vee) \xrightarrow{\text{forget}} \mathrm{Vect}$$

$$P_{L^+G}(\mathrm{Gr}) \xrightarrow{H^*} \mathrm{Vect}$$

$$\mathcal{F} \mapsto \bigoplus_i H^i(\mathrm{Gr}, \mathcal{F})$$

Note fibre functors are required to be a tensor functor, so we should check:

$$H^*(\mathcal{F} * \mathcal{G}) = H^*(\mathcal{F}) \otimes H^*(\mathcal{G})$$

This is because under the isomorphism:

$$\Omega_{\mathrm{pol}} K \simeq \mathrm{Gr}_G$$

then the convolution on Gr_G becomes usual convolution on the topological group on $\Omega_{\mathrm{pol}} K$:

$$\Omega_{\mathrm{pol}} K \times \Omega_{\mathrm{pol}} K \to \Omega_{\mathrm{pol}} K$$

$$(\mathcal{F}_1, \mathcal{F}_2) \mapsto m_{K*}(\mathcal{F}_1 \boxtimes \mathcal{F}_2)$$

Then we finish the check by Kunneth formula.

Recall that semi-infinite orbits $S_\nu = LN \cdot t^\nu \subset \mathrm{Gr}$, then $S_\nu \cap \mathrm{Gr}_{\leq\lambda}$ is non-empty iff $\nu \in \mathrm{Conv}(W \cdot \lambda)$ and has dimension $\langle \rho, \lambda + \nu \rangle$.

Corollary 8.1. *If $\mathcal{F} \in P_{L^+G}(\mathrm{Gr})$, then $H_c^*(S_\nu, \mathcal{F}|_{S_\nu})$ is nonzero in degree $\leq \langle 2\rho, \nu \rangle$.*

In fact, we have a more precise statement:

Theorem 8.2. *If $\mathcal{F} \in P_{L^+G}(\mathrm{Gr})$, then $H_c^*(S_\nu, \mathcal{F}|_{S_\nu})$ is concentrated in degree $\langle 2\rho, \nu \rangle$.*

Proof. Consider the oppositive orbit $T_\nu = LN^- \cdot t^\nu$. Then we have similar result:

$$T_\nu \cap \mathrm{Gr}_{\leq\lambda} \neq \emptyset \iff \nu \in \mathrm{Conv}(W \cdot \lambda)$$

and in which case $\dim(T_\nu \cap \mathrm{Gr}_{\leq\lambda}) = \langle \rho, \lambda - \nu \rangle$. The analogue of the corollary reads:

$$H^*(T_\nu, i_{T_\nu}^! \mathcal{F})$$

concentrate in degree $\geq \langle 2\rho, \nu \rangle$. The key ingredient of the proof is

Theorem (Hyperbolic Localization, Braden)**.** *Let X be a normal variety with a $\mathbb{G}_m$-action. Let $X^0 = X^{\mathbb{G}_m}$ be the fixed points. Now let:*

$$X^+ = \{x \in X \mid \lim_{z\to 0} z \cdot x \in X^0\}, \quad X^- = \{x \in X \mid \lim_{z\to\infty} z \cdot x \in X^0\}$$

Let:

$$X \xleftarrow{i^+} X^+ \xrightarrow{p^+} X^0, \quad X \xleftarrow{i^-} X^- \xrightarrow{p^-} X^0,$$

For $\mathcal{F} \in D_c^b(X)$, we have an isomorphism of $\mathbb{G}_m$-equivariant sheaves:

$$(p^+)_! (i^+)^* \mathcal{F} \simeq (p^-)_* (i^-)^! \mathcal{F}$$

In our case, let us consider $X = \mathrm{Gr}$, $X^0 = \{t^\nu\}$, $i^+ = i^- = \mathrm{id}$. We have:

$$H_c^*(S_\nu, \mathcal{F}|_{S_\nu}) \simeq ((p^+)_!\mathcal{F})_{t^\nu} \simeq ((p^-)_*\mathcal{F})_{t_\nu} \simeq H^*(T_\nu, i_{T_\nu}^! \mathcal{F})$$

again by proper base change. □

Corollary 8.3. *Let $\mathcal{F} \in P_{L^+G}(\mathrm{Gr})$, we have canonical decomposition:*

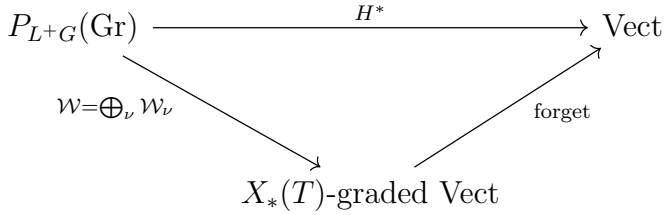
$$H^*(\mathrm{Gr}, \mathcal{F}) = \bigoplus_{\nu \in X_*(T)} H_c^{\langle 2\rho, \nu\rangle}(S_\nu, \mathcal{F}|_{S_\nu})$$

In other words:

$$H^n(\mathrm{Gr}, \mathcal{F}) = \bigoplus_{\langle 2\rho,\nu\rangle = n} H_c^n(S_\nu, \mathcal{F}) = H_c^n(\coprod S_\nu, \mathcal{F})$$

The idea is to filter Gr by $\langle 2\rho, \nu\rangle$, and the spectral sequence will degenerate due to some parity argument.

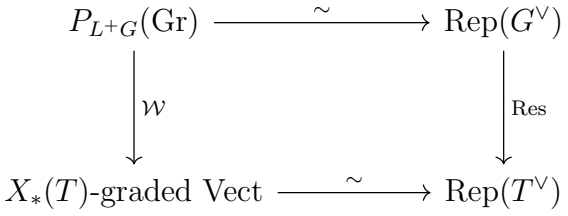
Now the fibre functor H^* factors through a more refined functor:



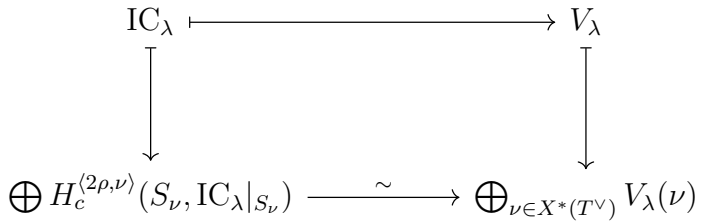
where $\mathcal{W}_\nu = H_c^{\langle 2\rho,\nu\rangle}(S_\nu, -)$. The functor $\mathcal{W}$ is called the *weight functor*.

Fact (MV): The weight functor $\mathcal{W}$ is also monoidal.

Remark. *The functor $\mathcal{W}$ is called the weight functor because under geometric Satake:*



In particular,



In fact, the space $H_c^{\langle 2\rho,\nu\rangle}(S_\nu, \mathrm{IC}_\lambda|_{S_\nu})$ has a basis given by MV cycles in $S_\nu \cap \mathrm{Gr}_{\leq\lambda}$.

The next goal is to explain the symmetric structure on $P_{L^+G}(\mathrm{Gr})$. The corresponding function theoretic statement is the fact that $\mathcal{H}_G^{\mathrm{sph}}$ is commutative: this is proved using Gelfand's trick $g \mapsto \sigma(g^{-1})$. We wish to upgrade this to a geometric argument.

The idea is that $g \mapsto \sigma(g^{-1})$ makes sense on LG, which induces an anti-involution on $P_{L^+G}(\mathrm{Gr})$ preserving IC_λ. Therefore, for each λ, μ we can obtain some isomorphism:

$$\mathrm{IC}_\lambda * \mathrm{IC}_\mu \simeq \mathrm{IC}_\mu * \mathrm{IC}_\lambda$$

But it is not clear how to make this functorial. To this end, we need to introduce *Drinfeld fusion*.

Recall that for $G = \mathrm{GL}_n$, we have:

$$\mathrm{Gr}_G(k) = \{\Lambda \subset F^n\}$$

which can be identified as a vector bundle $\mathcal{E}$ on $\mathcal{O}_F$ (choose a projective rank n $\mathcal{O}$-module), together with a trivialization on $\mathcal{E}|_{\mathrm{Spec}\ F}$ (choose a basis of F^n determining $\Lambda \hookrightarrow F^n$). But this makes sense for all G:

$$\mathrm{Gr}_G = \{(\mathcal{E}, \tau) \mid \mathcal{E} \in \mathrm{Bun}_G(\mathcal{O}_F), \tau : \mathcal{E}|_{\mathrm{Spec}\ F} \simeq G(F)\}$$

Now if X/k is a smooth curve, $x \in X(k)$, then:

$$\mathrm{Gr}_G \simeq \{(\mathcal{E}, \tau) \mid \mathcal{E} \in \mathrm{Bun}_G(X), \tau \text{ triv of } \mathcal{E}|_{X \setminus \{x\}}\}$$

The idea is to "move the point":

Definition 8.4 (Global / Beilinson-Drinfeld Gr)**.** In the above setting, we define:

$$\mathrm{Gr}_X = \{(x, \mathcal{E}, \tau) \mid x \in X, \mathcal{E} \in \mathrm{Bun}_G(X), \tau \text{ triv of } \mathcal{E}|_{X \setminus \{x\}}\}$$

$$\mathrm{Gr}_{X^2} = \{(x_1, x_2, \mathcal{E}, \tau) \mid x_1, x_2 \in X; \mathcal{E} \in \mathrm{Bun}_G(X); \tau \text{ triv of } \mathcal{E}|_{X \setminus \{x_1, x_2\}}\}$$

The upshot of this definition is that we have a (étale locally trivial) fibration:

$$\mathrm{Gr}_G \to \mathrm{Gr}_X \to X$$

The interesting behavior happens when we consider $\mathrm{Gr}_{X^2} \to X^2$. The fibre of (x_1, x_2) is:

1. $\mathrm{Gr} \times \mathrm{Gr}$, if $x_1 \neq x_2$;
2. Gr if $x_1 = x_2$.

This contrasts with finite-dimensional geometry, in which case the fibre dimension only jumps up. Now for λ_1, λ_2 dominant coweights, we can consider:

$$\mathrm{Gr}_{X, \leq \lambda_1} \hookrightarrow \mathrm{Gr}_X$$

by restricting relative position constraint on the lattice determined by $(\mathcal{E}, \tau)$. Now observe that $\mathrm{Gr}_{X^2}|_{X^2 \setminus \Delta X} \simeq \mathrm{Gr}_X \times \mathrm{Gr}_X\ |_{X^2 \setminus \Delta X}$. So let's consider:

$$\mathrm{Gr}_{X^2, \leq (\lambda_1, \lambda_2)} = \overline{(\mathrm{Gr}_{X, \leq \lambda_1} \times \mathrm{Gr}_{X, \leq \lambda_2})|_{X^2 \setminus \Delta}}$$

It turns out that we will have a diagram:

$$\begin{array}{ccccc}
\mathrm{Gr}_{\leq\lambda_1+\lambda_2} & \hookrightarrow & \mathrm{Gr}_{X^2,\leq(\lambda_1,\lambda_2)} & \hookleftarrow & \mathrm{Gr}_{\leq\lambda_1}\times\mathrm{Gr}_{\leq\lambda_2} \\
\downarrow & & \downarrow & & \downarrow \\
(x,x) & \hookrightarrow & X^2 & \hookleftarrow & (x_1,x_2)
\end{array}$$

Definition 8.5 (Fusion Product)**.** Given $\mathcal{F}_1, \mathcal{F}_2 \in P_{L^+G}(\mathrm{Gr})$, we can "spread out" $\mathcal{F}_1 \boxtimes \mathcal{F}_2$ to $X^2 \setminus \Delta X$ to get a sheaf $(\mathcal{F}_1 \boxtimes \mathcal{F}_2)_{X^2\setminus\Delta X}$ on $X^2 \setminus \Delta X$ (we use the L^+G-equivariance property to kill the ambiguity of identifying $G(\widehat{\mathcal{O}_X}) \simeq G(k[[t]])$). Now we use the general machinery of *nearby cycle functor* to obtain the fusion product:

$$\mathcal{F}_1 \circledast \mathcal{F}_2 = \Psi_{\Delta X}(\mathcal{F}_1 \boxtimes \mathcal{F}_2)_{X^2\setminus\Delta X}|_{(x,x)} \in P_{L^+G}(\mathrm{Gr})$$

This will independent of the choice of $x \in X$.

It turns out $\circledast$ agrees with convolution product $*$, but has the advantage of having a symmetric structure from birth.

To see the agreement, let's consider:

$$\widetilde{\mathrm{Gr}}_{X^2} = \{(x_1, x_2, \mathcal{E}_{\mathsf{triv}} \dashrightarrow \mathcal{E}' \dashrightarrow \mathcal{E})\}$$

where the first arrow is an isomorphism over $X \setminus x_1$, and the second arrow is an isomorphism over $X \setminus x_2$. Now we have a natural map via forgetting $\mathcal{E}'$:

$$v : \widetilde{\mathrm{Gr}}_{X^2} \to \mathrm{Gr}_{X^2}$$

This is an isomorphism over $X^2 \setminus \Delta X$. The fibre at ΔX is:

$$v|_{(x,x)} \colon \{\mathcal{E}_{\mathsf{triv}} \dashrightarrow \mathcal{E}' \dashrightarrow \mathcal{E}\} \to \mathrm{Gr}$$

by forgetting $\mathcal{E}'$. Note that LHS is naturally isomorphic to:

$$LG \times^{L^+G} \mathrm{Gr}$$

and the forgetting map becomes just the action map $m : LG \times^{L^+G} \mathrm{Gr} \to \mathrm{Gr}$. Now we can spread out $\mathcal{F}_1, \mathcal{F}_2 \in P_{L^+G}(\mathrm{Gr})$ to a sheaf $\mathcal{F}_1 \widetilde{\boxtimes} \mathcal{F}_2 \in D^b(\widetilde{\mathrm{Gr}}_{X^2})$. This has the property that:

$$v_*(\mathcal{F}_1 \widetilde{\boxtimes} \mathcal{F}_2)|_{X^2\setminus\Delta X} = (\mathcal{F}_1 \boxtimes \mathcal{F}_2)_{X^2\setminus\Delta X}$$

because v is just identity outside the diagonal. Now since v is proper, and nearby cycle functor commutes with proper pushforward, we have:

$$\mathcal{F}_1 \circledast \mathcal{F}_2 = \Psi_{\Delta X}(\mathcal{F}_1 \boxtimes \mathcal{F}_2)_{X^2\setminus\Delta X}|_{(x,x)} = v_* \circ \Psi_{\Delta X}(\mathcal{F}_1 \widetilde{\boxtimes} \mathcal{F}_2)_{X^2\setminus\Delta X}|_{(x,x)}$$

Since $v = m$, it suffices to show:

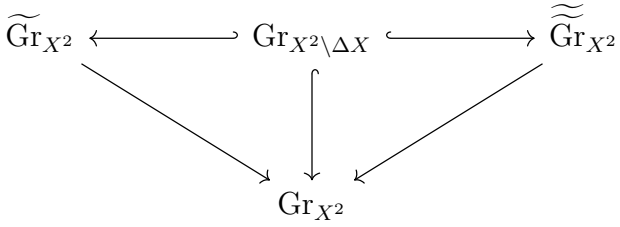
$$\Psi_{\Delta X}(\mathcal{F}_1 \widetilde{\boxtimes} \mathcal{F}_2)_{X^2\setminus\Delta X}|_{(x,x)} = p_1^*\mathcal{F}_1 \otimes p_2^*\mathcal{F}_2 \in D^b(LG \times^{L^+G} \mathrm{Gr})$$

But this equality holds generically (outside the special fibre), so we expect after passing through the nearby cycle it still holds.

Now let's explain the commutativity. Note the definition of $\widetilde{\mathrm{Gr}_{X^2}}$ is asymmetric, so we should also consider:

$$\widetilde{\widetilde{\mathrm{Gr}}}_{X^2} = \{(x_1, x_2, \mathcal{E}_{\mathsf{triv}} \dashrightarrow \mathcal{E}' \dashrightarrow \mathcal{E})\}$$

where the first arrow is isomorphism outside x_2 and the second arrow is isomorphism outside x_1. Now we have diagram:



But the whole construction of $\circledast$ only involves spreading out on $X^2 \setminus \Delta X$, we see that:

$$\mathcal{F}_1 * \mathcal{F}_2 \simeq \mathcal{F}_1 \circledast \mathcal{F}_2 \simeq \mathcal{F}_2 * \mathcal{F}_1$$

Remark. *The Drinfeld's fusion $\circledast$ is absolutely a* **new structure***: for example, it captures the functoriality of the isomorphisms $\mathcal{F}_1 * \mathcal{F}_2 \simeq \mathcal{F}_2 * \mathcal{F}_1$.*

Let's briefly discuss on how to finish the proof. We need to break into the following steps:

1. Show that the fibre functor functor $P_{L^+G}(\mathrm{Gr}) \xrightarrow{H^*}$ Vect is exact, symmetric monoidal functor.

2. Consider $\mathcal{G} = \mathrm{Aut}^{\otimes}(H^*)$, this is an algebraic group over $\overline{\mathbb{Q}_\ell}$ via Tannakian reconstruction. We need to check:

   Theorem 8.6 (MV)**.** $\mathcal{G} \simeq G^\vee$

   where $G^\vee$ is combinatorially defined Langlands dual group (only defined up to isomorphism).

3. Consider the map:

   $$P_{L^+G}(\mathrm{Gr}) \xrightarrow{\oplus H^i} \mathbb{Z}\text{-graded Vect}$$

   This corresponds to a tensor functor:

   $$\mathrm{Rep}(G^\vee) \xrightarrow{\otimes} \mathrm{Rep}(\mathbb{G}_m)$$

   This should come from $\mathbb{G}_m \xrightarrow{2\rho} G^\vee$. So we actually equip $G^\vee$ with:

   $$\mathbb{G}_m \xrightarrow{2\rho} T^\vee \hookrightarrow G^\vee$$

   the second map corresponds to weight functor $\mathrm{Rep}(G^\vee) \to \mathrm{Rep}(T^\vee)$. Moreover, $G^\vee$ has a *canonical pinning* (from c_1 on Gr).

We can interpret the representation theoretic information in terms of $P_{L^+G}(\mathrm{Gr})$. One question in this direction in the following: given $H^\vee \subset G^\vee$, we have restriction of scalar:

$$\mathrm{Rep}(G^\vee) \to \mathrm{Rep}(H^\vee)$$

What is the corresponding tensor functor:

$$P_{L^+G}(\mathrm{Gr}_G) \xrightarrow{\otimes} P_{L^+H}(\mathrm{Gr}_H)$$

For example, if $H = T$ we're just getting MV cycles.

9 A Glimpse into Geometric Langlands

Let's denote

$$\mathrm{Sat} : \mathrm{Rep}(G^\vee) \xrightarrow{\sim} \mathrm{Perv}(L^+G \setminus \mathrm{Gr})$$

So that each representation $V \in \mathrm{Rep}(G^\vee)$ corresponds to some spherical perverse sheaf $\mathrm{Sat}(V)$ on Gr.

Now we fix a curve $X/\mathbb{C}$. Let's consider:

$$\mathrm{Hk}_{G,X} = \{(x, \mathcal{E}_1, \mathcal{E}_2, \tau) \mid x \in X, \mathcal{E}_1 \in \mathrm{Bun}_G(X), \mathcal{E}_2 \in \mathrm{Bun}_G(X), \tau : \mathcal{E}_1|_{X\setminus\{x\}} \to \mathcal{E}_2|_{X\setminus\{x\}}\}$$

Then this naturally sits inside:

$$\begin{array}{ccccc} \mathrm{Bun}_G(X) & \xleftarrow{h_1} & \mathrm{Hk}_{G,X} & \xrightarrow{h_2} & \mathrm{Bun}_G(X) \\ & & \downarrow{\scriptstyle s} & & \\ & & X & & \end{array}$$

Use this diagram we can define a functor for each $V \in \mathrm{Rep}(G^\vee)$ roughly as:

$$\mathsf{H}_V : D(\mathrm{Bun}_G(X)) \to D(X \times \mathrm{Bun}_G(X))$$

$$\mathcal{F} \mapsto (s, h_2)_!(h_1^*\mathcal{F} \otimes \mathrm{Sat}(V))$$

This makes sense because we saw we can spread on $\mathrm{Sat}(V)$ to a sheaf on $\mathrm{Hk}_{G,X}$. Also note that this operator H_V is nothing but convolve with integral kernel given by V.

Now we're in the right place to talk about Hecke eigensheaves. We say $\mathcal{F} \in D(\mathrm{Bun}_G)$ is a *Hecke eigensheaf* with *eigenvalue* $E \in \mathrm{LocSys}_{G^\vee}(X)$ if for every $V \in \mathrm{Rep}(G^\vee)$:

$$\mathsf{H}_V(\mathcal{F}) = E_V \boxtimes \mathcal{F}$$

for $E_V = E \times^{G^\vee} V$.

There's a similar construction on the level of $\mathcal{D}$-modules, for I can use Riemann-Hilbert to view $\mathrm{Sat}(V)$ as a $\mathcal{D}$-module on $\mathrm{Hk}_{G,X}$. Now the geometric Langlands correspondence roughly reads:

$$\mathcal{D}\text{-mod}(\mathrm{Bun}_G(X)) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{G^\vee}(X))$$

The Hecke-eigen-objects on both sides will play an important role for establishing it. Moreover, the equivalence will be be compatible with the Hecke actions on both sides.